

Track 21: Raspberry Pi

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Linux on Pi → GPIO → Python → Sensors → Camera → Networking → IoT Projects → Edge AI

Related: [Track 19 — Robotics & IoT](#) | [Track 26 — Computer Vision](#)

Track Overview

Who this is for: Makers, IoT developers, and students building edge computing and smart device projects.

Prerequisites: Basic Python; Linux command line taught from Week 1.

Tools: Raspberry Pi 4/5, Raspberry Pi OS, GPIO library (gpiozero/RPI.GPIO), Pi Camera, MQTT, Flask/FastAPI.

Hardware: Raspberry Pi board + power supply + microSD + sensor kit

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Pi awareness	GPIO automation project
1 Month	80–100	Junior Pi developer	Smart home hub
3 Months	250–300	Internship-ready	IoT edge server capstone
6 Months	500–600	Edge developer	Camera + AI edge project
9 Months	750–900	Advanced edge engineer	Multi-node Pi cluster + AI

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Pi setup, Linux CLI, SSH, VNC	Configured Pi headless
D3–4	GPIO with Python, LEDs, buttons	GPIO control scripts
D5–6	Sensors (DHT, PIR), data logging	Sensor logger to CSV
D7–8	Flask web server, REST API on Pi	Local web dashboard
D9–10	Capstone	Smart monitoring station on Pi

1-Month Foundation

Week	Topics	Project
W1	Pi OS, Python, GPIO, systemd services	Auto-start GPIO service
W2	Sensors, SQLite, scheduling (cron)	Data logging server
W3	MQTT, cloud sync, remote control	IoT hub with MQTT
W4	Pi Camera, streaming basics	Security camera / timelapse project

3-Month Job Ready

Month	Topics	Projects
M1	Advanced GPIO, I2C/SPI sensors	Sensor array project
M2	Flask/FastAPI, Docker on Pi	Edge API server
M3	Kiosk mode, automation, capstone	Smart agriculture / home automation hub

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	OpenCV on Pi, edge inference (TFLite)	Object detection on Pi
M5	Clustering (multiple Pis), load balancing	2-node Pi cluster
M6	Capstone	Edge AI camera system

9-Month Advanced

Capstones: Edge AI surveillance system + distributed Pi IoT network | **Level:** L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	Linux, GPIO, Python, networking
Lab Completion	40%	Scripts run; services start on boot
Project	30%	Reliability, documentation, demo
Portfolio	10%	GitHub, wiring photos, videos

Appendix: Mentor Notes

Use quality power supply (5V 3A+). Image SD cards before major labs. Cross-track with [Track 26](#) for camera/AI depth.